

# Software Engineering Lab 1

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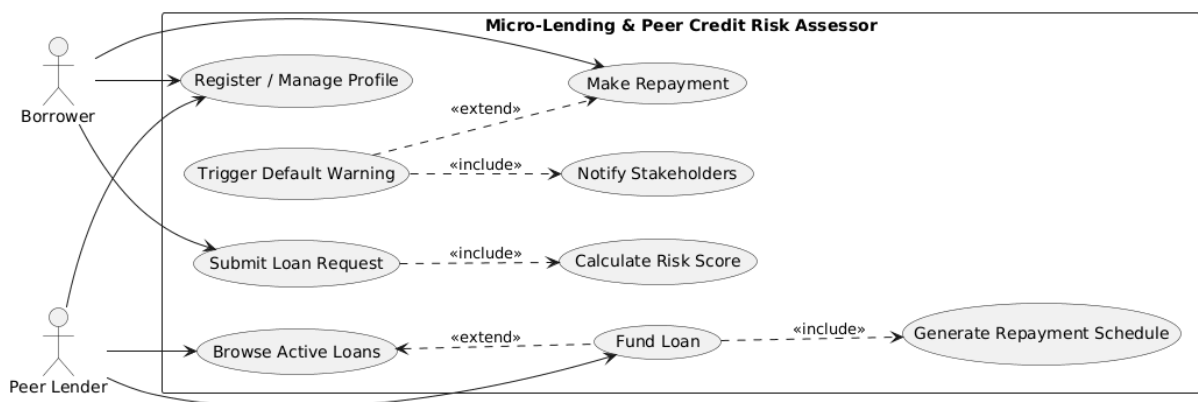
Class: 5B

## Deliverable 1: Requirements Table

| ID     | Type       | Description                                                                                                                                  | Priority | Acceptance Criteria                                                                                                                                                     | Rationale                                                     |
|--------|------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| FR-001 | Functional | The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators. | High     | Pass: Risk tier (Low/Med/High) accurately assigned; Fail: Incomplete financial profile generates credit approval.                                                       | Essential for peer lenders to gauge risk before funding.      |
| FR-002 | Functional | The system shall generate structured repayment schedules for funded loans.                                                                   | High     | Pass: Accurate EMI schedule with dates is generated upon loan approval; Fail: Schedule is missing, has wrong dates, or EMIs do not add up to principal + interest.      | Borrowers need clear timelines to prevent defaults.           |
| FR-003 | Functional | The system shall trigger automated default warnings to borrowers and lenders when an EMI is missed.                                          | Med      | Pass: Warning notification sent at exactly T+1 days past the due date; Fail: No warning is sent, or it is sent before the due date has passed.                          | Proactive risk management and communication.                  |
| FR-004 | Functional | Borrowers shall be able to submit new loan requests specifying the required amount and purpose.                                              | High     | Pass: Loan request is successfully saved and queued for risk assessment; Fail: A request with a missing amount or purpose is accepted, or a valid request is not saved. | Core platform functionality for initiating the lending cycle. |

| ID             | Type                   | Description                                                                                                                 | Priority | Acceptance Criteria                                                                                                                                                          | Rationale                                                    |
|----------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>FR-005</b>  | Functional             | Peer Lenders shall be able to browse and fund active loan requests based on borrower risk scores.                           | High     | Pass: Lender funds are deducted and the loan is marked as "Funded"; Fail: The loan is marked Funded without funds being deducted, or funds are deducted twice.               | Enables the peer-to-peer transaction aspect of the platform. |
| <b>NFR-001</b> | Performance & Security | Loan repayment transactions and EMI distribution must execute inside database ACID transactions to prevent double-spending. | High     | Pass: A simulated failure mid-repayment rolls back completely and concurrent repayment tests show no duplicate debits; Fail: A partial or duplicate transaction is recorded. | Prevents severe financial data corruption.                   |
| <b>NFR-002</b> | Performance            | The platform's risk assessment and repayment dashboard must load within 2 seconds.                                          | Med      | Pass: Automated load testing confirms page load times $\leq 2000$ ms; Fail: The dashboard takes longer than 2000 ms to load under normal load.                               | Ensures a smooth user experience for stakeholders.           |

## Deliverable 2: UML Use-Case Diagram



## Deliverable 3: Use-Case Flow Specification

**Use Case Name:** Calculate Risk Score (Compute Borrower Credit Risk Rating)

**1. Use Case Description** This use case (included by "Submit Loan Request") describes the process by which the Micro-Lending platform evaluates a borrower's financial profile to calculate and assign a credit risk tier (Low, Medium, or High).

### 2. Actors

- **Primary Actor:** Borrower
- **Secondary Actor:** Peer Lender (views the assigned risk tier)

### 3. Preconditions

- The Borrower has successfully registered an account on the platform.
- The Borrower has initiated a new loan request and populated their complete financial profile, including income, active debts, and employment history.

### 4. Postconditions

- **Success:** The Borrower is assigned a verified Risk Tier (Low/Med/High), which is saved to the database and made visible to potential Peer Lenders.
- **Failure:** The risk assessment is aborted, no score is assigned, and the Borrower is prompted to complete their profile.

### 5. Main Success Scenario (Standard Flow)

1. The Borrower submits a loan request.
2. The system triggers the Risk Assessor Engine.
3. The system retrieves the borrower's submitted financial data: debt-to-income ratio, repayment history, and employment stability indicators.
4. The heuristic rules engine evaluates the retrieved indicators against the platform's risk baseline.
5. The system successfully calculates a numerical risk score.
6. The system maps the numerical score to the corresponding Risk Tier (Low, Med, or High).
7. The system updates the loan request status to "Assessed" and displays the tier to both the Borrower and Peer Lenders.

### 6. Alternate Flow 1 (Incomplete Financial Profile)

- 2a. During step 3 of the Main Success Scenario, the system detects missing mandatory data (e.g., missing employment stability indicators).
- 2b. The system halts the risk calculation process immediately.
- 2c. The system generates an error notification: "Incomplete financial profile. Please update your employment history to proceed."
- 2d. The credit approval fails, and the loan request remains in the "Draft" state until the Borrower updates the required fields.